

Reconstruction-Based Supervised Contrastive Learning for Unknown Device Identification in Nonintrusive Load Monitoring

Yinghua Han¹, Haoqi Chen¹, Jingrun Wu¹, and Qiang Zhao¹

Abstract—Nonintrusive load monitoring (NILM) involves deploying monitoring devices at the customer's bus entrance to decompose the total power captured into individual operational data for each device. Existing NILM load identification algorithms primarily adopt closed-set identification, where the same types of devices are used for both training and testing. However, a more realistic scenario is open-set identification, where device types in the training and test sets differ, encompassing not only devices from the training set but also unknown devices. To address this challenge, we propose a method leveraging reconstruction and supervised contrastive learning based on voltage–current (V – I) trajectories. Supervised contrastive learning improves the coherence of V – I trajectories within identical appliance classes, facilitating the model's precise comprehension and emulation of distinctive appliance behavioral patterns. By introducing an autoencoder (AE) into the model, associating each appliance class with its unique subset of features. Known V – I trajectories activate associated features, preserving the topological characteristics of the trajectory. In contrast, unknown V – I trajectories lack reconstruction mappings, resulting in thoroughly mutated trajectories. Moreover, we innovate data augmentation for contrastive learning by introducing a denoising AE (DAE). This approach introduces noise in V – I trajectories, the model learns to remove noise while preserving real appliance characteristics, generating more robust V – I trajectories. Experimental results on two publicly available datasets demonstrate the method's outstanding performance.

Index Terms—Autoencoder (AE), nonintrusive load monitoring (NILM), supervised contrastive learning, unknown devices detection, voltage–current (V – I) trajectory.

I. INTRODUCTION

ENERGY is one of the driving forces of global development and life, underpinning every aspect of modern society, from infrastructure to production and everyday life.

Manuscript received 23 October 2023; revised 25 December 2023; accepted 30 January 2024. Date of publication 7 March 2024; date of current version 19 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U21A20475 and in part by the Fundamental Research Funds for the Central Universities under Grant N2223001. The Associate Editor coordinating the review process was Dr. Zhenbing Zhao. (Corresponding author: Haoqi Chen.)

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Digital Object Identifier 10.1109/TIM.2024.3374287

However, as the world's population grows and industrialization accelerates, the energy demand is increasing rapidly, which raises a complex set of challenges, including energy security, sustainability, and environmental impacts. Smart grids, as a necessary solution for innovative energy management systems, are a key enabler for the future of smart energy consumption [1], [2]. For example, in the United States (USA), where primary energy and buildings consume more than 38% and 76% of electricity, respectively, reductions of up to 15%–40% can be achieved using whole-building energy management systems [3]. The continued deployment of smart meters and different commercial devices makes it possible to perform load monitoring in buildings and homes based on a single measurement of current (and sometimes even voltage).

Load monitoring is categorized into intrusive load monitoring (ILM) and nonintrusive load monitoring (NILM) [4], [5]. However, ILM methods usually require the installation of specialized sensors or hardware, which may involve intrusive modifications to existing systems. To address these issues, NILM was developed to provide a more flexible, cost-effective, and nonintrusive way to monitor loads in real-time.

NILM was first proposed by Hart [5]. NILM contains two main core tasks: decomposition and identification. The goal of the decomposition task is to separate the power consumption signal of each electrical device from the overall aggregated signal. It has been widely developed in the last decades due to advances in the field of machine learning and deep learning algorithms. Early NILM research was mainly based on hidden Markov models (HMMs), which are commonly used for probabilistic modeling of time series data [6]. Kong et al. [7] proposed a hierarchical HMM framework to model home appliances to obtain feature identifiers for multimode electrical devices to improve power decomposition. Egarter et al. [8] proposed modeling household consumption using a factorial HMM with particle filtering to estimate device states. The development of deep learning solutions introduced a neural network approach in the NILM environment, which quickly became the basis for state-of-the-art implementations that outperformed previous research. Chen et al. [9] proposed self-supervised learning, which does not require labeled device-level data from the target dataset or houses. The research presented in [10] had a significant impact on the NILM literature. The authors proposed a new sequence-to-point model that was trained to predict the mid-point of the output sequence rather than the entire sequence. Xiong et al. [11] preprocessed the data using discrete wavelet

transform. The processed data is transferred to an enhanced sequence-to-point network for training.

Identification tasks have traditionally been conceptualized as classification tasks, allowing for the exploration of algorithms and models applicable to classification. Machine learning's success in image processing suggests that voltage–current (V – I) trajectory signatures could bridge the gap between computer vision and NILM [12]. Lam et al. [13] pioneered the concept of V – I trajectory features, derived from plotting a device's operating current against voltage. Load features, such as area, symmetry, and cyclic direction, were then extracted from the V – I trajectory. Liu et al. [14] introduced a transfer learning method leveraging V – I trajectory data to address challenges associated with limited data label acquisition. de Paula Rodrigues and da Silveira [15] proposed two steady-state features derived from the curvature scale space of the V – I trajectory, aiming to enhance the image representation of V – I trajectories by incorporating structural elements related to their general shape, along with features overlooked during their generation. Han et al. [16] suggested the application of a semi-supervised smoothed teacher graph learning network for V – I trajectory recognition. Zhang et al. [17] introduced a method involving empirical modal decomposition (EMD) to filter sampled data and applied deep learning for automatic feature extraction from constructed V – I trajectories.

While the aforementioned approaches for energy decomposition and load identification exhibit commendable performance in closed-set scenarios, it is imperative to acknowledge that NILM inherently poses a nonsmooth problem [18]. The real-world scenario deviates from a consistent alignment of appliance types between training and testing [19], [20]. In practical terms, it is more realistic to encounter incomplete appliance types during training and introduce unknown appliances during testing, expecting the model to adeptly identify them. Therefore, when faced with unknown appliances, the model should demonstrate proficiency in their recognition [12]. Regrettably, the aforementioned methods may erroneously categorize an unknown appliance as a known one in such encounters.

To tackle the issue of unknown devices and enhance the applicability of NILM in real-world scenarios, researchers have devised algorithms focused on the detection of unknown devices. In the pursuit of anomaly detection using NILM, as discussed in [21], an algorithm was proposed to identify abnormal behaviors in appliances like refrigerators and air conditioners. Another noteworthy algorithm, outlined in [22], not only discerned open devices and their power consumption levels but also effectively detected the presence of unlearned unknown devices. Zhao et al. [23] introduced a deep pairwise supervised hashing method tailored for V – I trajectories. This method utilized distinct hash codes to differentiate known devices, generating novel codes for unknown devices and subsequently identifying them based on Hamming distance. Yu et al. [24] leveraged a Siamese network to compute V – I trajectory similarity, achieving load recognition by matching signatures to a feature library. The identification of unknown devices was facilitated by dynamically adding new features

to the library. In [25], the potential presence of a new device was considered if the model's overall performance dropped below a predetermined threshold. An innovative approach proposed in [26] involved mapping V – I trajectories to a new feature space using a Siamese network, followed by the assignment of device samples to clusters or marking them as unidentified using the DBSCAN clustering algorithm. Han et al. [27] introduced a conditional generative adversarial network that integrated a variational AE (VAE) and a capsule network. This network adeptly identified known devices and demonstrated proficiency in detecting unknown devices. Despite these advancements, these approaches often overlook the challenge of data imbalance in real-world scenarios, where appliances exhibit diverse usage patterns. For instance, a refrigerator may run continuously, while a washing machine might be used intermittently. Most deep learning algorithms depend on large and balanced datasets, and their training efficiency is constrained, potentially leading to overfitting and reduced accuracy [28]. Consequently, there is a compelling need for an algorithm capable of unknown device detection while effectively addressing the data imbalance problem.

For the unknown device detection problem, similar to the open-set identification and anomaly detection problems, the V – I trajectories of unknown devices can be considered anomalous data [29]. The rise of contrastive learning in recent years has also driven the development of open-set identification and anomaly detection. Yang et al. [30] proposed a new open-set classifier that created a compact representation space to reduce false negatives using a supervised contrast learning approach and proposed an integration-based out-of-distribution detection module to reduce false positives at the pixel and semantic level. In [31], the effectiveness of supervised contrastive learning for open-set recognition with fewer parameters was demonstrated. A method was proposed to compute the anomaly scores for each feature separately and sum them up in a weighted manner instead of concatenating them. Vaze et al. [32] first established a strong baseline by taking state-of-the-art algorithms from new category discovery and adapting them for this task. Next, a visual converter with supervised contrastive learning was used in an open-world environment. Then, unlabeled data was automatically clustered into visible and invisible classes using a semi-supervised K -means approach. Methods utilizing reconstruction in open-set identification and anomaly detection were also widely used. Huang et al. [33] proposed a class-specific semantic reconstruction (CSSR) approach that integrated the power of AE and prototype learning. CSSR replaced prototype points with manifolds represented by class-specific AEs, then modeled each known class on a single manifold, and measured class belongingness through AE's reconstruction error. Oza and Patel [34] proposed an open-set recognition algorithm for class conditional AE. The algorithm divided the training process into two subtasks: closed-set classification and open-set recognition. The encoder learned the first task by closed-set classification training, while the decoder learned the second task by reconstructing conditions on class identity. Kawachi et al. [35] proposed a supervised anomaly detection

algorithm based on a VAE. The proposed method could detect known anomalies and unknown anomalies in a better way compared to the traditional unsupervised VAE.

Referring to these methods above, we apply supervised contrastive learning and a reconstruction-based approach to NILM unknown appliance detection. In load recognition, similar $V-I$ trajectories occur for different kinds of appliances, e.g., compact fluorescent lamps (CFLs) and laptops. Such similar $V-I$ trajectories create a great deal of trouble for unknown appliance recognition. Contrastive learning is a combination of deep learning hierarchical learning features and custom sample distances to handle data imbalance. Supervised contrastive learning maximizes the consistency between the $V-I$ trajectories of appliances of the same class while excluding other kinds of appliances in the batch, and it compresses the $V-I$ trajectories into potential spatial representations, enabling the model to learn various properties of the $V-I$ trajectories to facilitate a better distinction between similar $V-I$ trajectories. When we train the model to maximize the vectors between similar $V-I$ trajectories, the model learns the clustering of similar data points in the latent space, which allows the model to distinguish between similar $V-I$ trajectories. Although there are previous successful examples of supervised contrastive learning, such as open-set recognition [31] and intelligent fault diagnosis [36], this study is the first to apply supervised contrastive learning to device detection. All of the above methods using reconstruction identify unknown categories by calculating the reconstruction error. Since the AE (or VAE) has learned images of known categories during training, then images originating from unknown categories generate high reconstruction errors. However, AE is not robust to noise and impairs the performance of open and closed sets when the input image contains redundant information [33], [37], [38]. In the real world, data collected by sensors may be affected by noise and outliers [39], [40]. This can lead to instability in AE-based methods. Therefore, instead of calculating the reconstruction error, we only seek to maximize the reduction of the known $V-I$ trajectories, while the unknown electrical $V-I$ trajectories will have a large degree of variation. The difference between known and unknown can be expanded using this method [33]. We look at these two aspects to improve the sensitivity of the algorithm to unknown appliances and ensure that the known appliances can also be identified with a high degree of accuracy.

In this article, we introduce a novel algorithm for detecting unknown devices, enhancing the NILM system's ability to identify both known and unknown devices. Our approach's primary contributions can be summarized in four key aspects as follows.

- 1) A new unknown device detection model is proposed, which combines supervised contrastive learning and an AE to effectively detect unknown devices and classify known devices.
- 2) We pretrained AE with known $V-I$ trajectories and inserted them when fine-tuning the classifier to reconstruct the semantic representation of the $V-I$ trajectories, increasing the discrepancy between known and unknown appliances.

- 3) Improved the data augmentation method by DAE on the $V-I$ trajectory after adding random noise to obtain an image that is different from the original $V-I$ trajectory, yet maximizes the preservation of appliance information.
- 4) We evaluated our detection methodology on two publicly accessible datasets, and the experimental findings substantiate the effectiveness and superiority of our proposed method within the academic context.

In the subsequent sections of this article, we structure our content as follows. In Section II, we provide an elaborate account of both the problem statement and proposed approach. Moving on to Section III, we focus on the experimental design and conduct a comprehensive discussion of the results obtained. Concluding our article in Section IV, we offer a concise summary and outline potential directions for future research.

II. PROBLEM STATEMENT AND PROPOSED APPROACH

In this section, we begin by outlining the process of extracting appliance $V-I$ trajectories from time series voltage and current data. We then introduce the challenge of detecting unknown appliances. Finally, we provide a comprehensive overview of our proposed model, covering key aspects such as the supervised comparison loss and its derivation process.

A. $V-I$ Trajectory Feature Extraction

NILM begins by acquiring the user's voltage and current measurements or total power consumption data, followed by the extraction of corresponding feature codes. Device switching events are detected to gather device signatures for training the NILM model. Utilizing $V-I$ trajectory features, a classification algorithm effectively identifies these switching events. To extract each device's $V-I$ trajectory, we assume a single device undergoes a state change at a specific time point. We collect total voltage and current values for T complete cycles once the device stabilizes, denoted as V_{on} , V_{off} , I_{on} , and I_{off} , respectively. The appliance's steady-state voltage and current at the switching event can be computed as $(v_{on} + v_{off})/2$ and $I_{off} - I_{on}$. Using load voltage and current data, we generate a 2-D $V-I$ trajectory image. In this article, to increase the number of features in the $V-I$ trajectory to improve the distinguishability of its $V-I$ trajectories from different devices, Hue, saturation, value (HSV) color space is used to color-code the $V-I$ trajectory to enhance the uniqueness of the load features, which is then used to train the classifier to have identified the appliances [14].

B. Unknown Devices Detection

In a real-world NILM application scenario, the appliances used by consumers are not static; one or more new appliances are added according to their needs, and various malfunctioning or inefficient appliances are eliminated. When an existing appliance is removed, we can assume that it is turned off and does not negatively impact overall NILM. However, most of the existing methods do not take this into account and when new appliances appear, they may be incorrectly classified as one of the categories of known appliances,

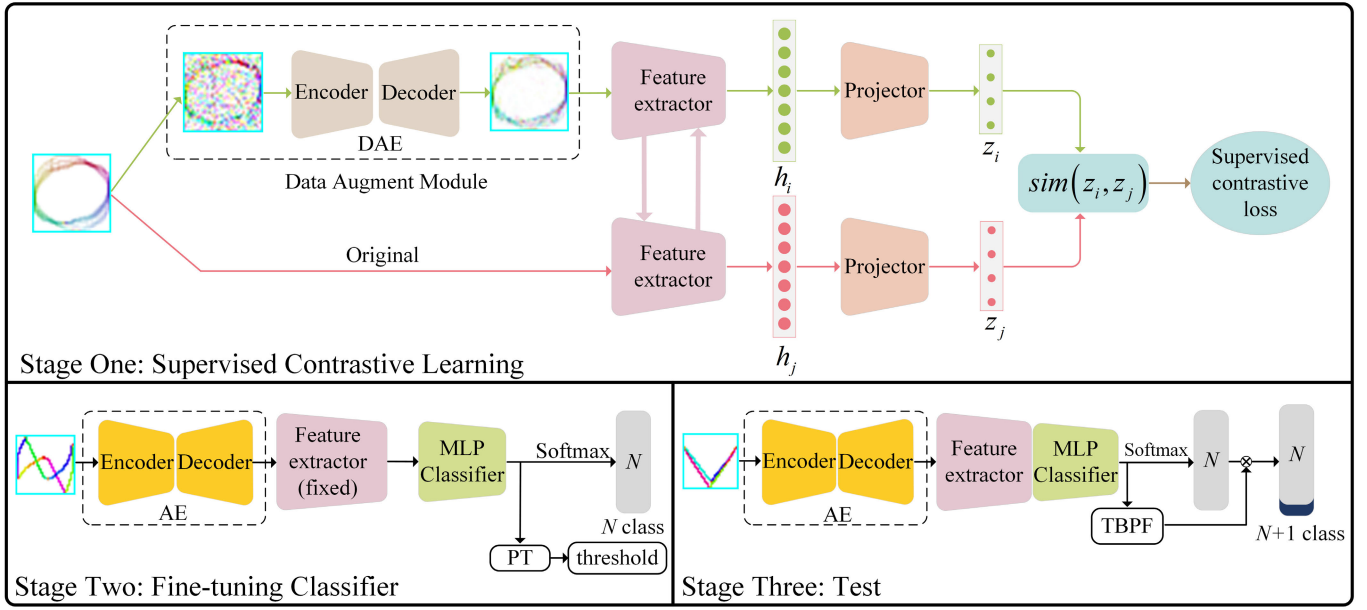


Fig. 1. Pipeline of the proposed model for unknown devices detection.

which has an impact on the identification task in NILM. Therefore, the NILM algorithm needs to both identify known devices and be able to detect unknown devices. This is an $N + 1$ class classification problem, where N stands for N known devices and 1 stands for unknown devices. Appliances that are present during testing and do not have their corresponding $V-I$ trajectory during training are uniformly classified as unknown devices.

C. Proposed Model Architecture

As shown in Fig. 1, our model is a three-stage process that uses contrast learning to train and test the network. In the first stage of the supervised contrastive learning process, we improve the data augmentation approach to better fit the unknown device detection problem in NILM. Instead of using the traditional methods of cropping, rotating, and brightness changing, a better feature extractor is obtained by denoising the noise-added $V-I$ trajectory to get an image that retains more device information but is not the same as the original $V-I$ trajectory. The transformed $V-I$ trajectory and the original $V-I$ trajectory are fed into the supervised contrastive learning network to compute the supervised contrastive loss. In the second stage, we freeze and transfer the pretrained feature extractors from the first stage and discard the projection network. Then, a multilayer perceptron (MLP) is connected on top of the frozen feature extractor and trained. Consider that some device components may operate on the same or similar principles as other classes of devices, resulting in similarities in the $V-I$ trajectories of different devices. To mitigate the resemblance between $V-I$ trajectories of known and unknown devices, we incorporate a pretrained AE initialized with $V-I$ trajectories of known devices as an upper layer of the feature extractor. When the $V-I$ trajectories of known devices pass through the AE, they can be well reproduced, and the $V-I$ trajectories of unknown devices pass through the AE with

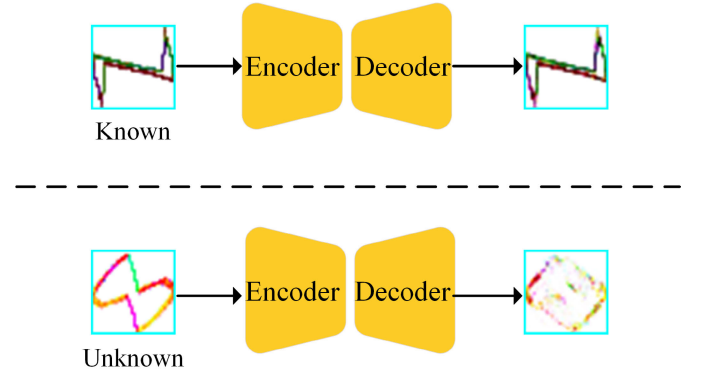


Fig. 2. Comparison of known and unknown $V-I$ trajectory through the AE.

a larger gap from the original image, so that they can be better differentiated between the known devices and the unknown devices, and a visual explanation is provided in Fig. 2. In the third stage, the $V-I$ trajectories of the known and unknown devices are fed into the model together, which is used to test the performance of the model. Next, we delve into the detailed architecture of the model, breaking it down step by step.

- 1) *Data Augmentation Module, Aug($\cdot$):* Our data augmentation module differs from the approach used in traditional data augmentation in that we use a DAE to recover the $V-I$ trajectory x after adding random noise to obtain x' . Previous methods of data augmentation were achieved by rotation, cropping, and brightness and color changes. However, these methods are not fully applicable to the $V-I$ trajectory. For example, cropping will intercept a certain part of the image, but this may make the $V-I$ trajectory intercepted on different appliances similar. It loses information about features such as asymmetry, loop direction, and size of the enclosed region area contained in the shape. Color and brightness changes are also problematic. The $V-I$ trajectories we

used are HSV color-coded. The color contains characteristic information such as trajectory motion information, active and reactive power ratios, and trajectory repeatability. When the color and brightness are changed, the corresponding electrical features are also changed. Some other data augmentation methods have similar problems. Therefore, we propose to use DAE for data augmentation [41]. This preserves the color-coded additional appliance features without drastically changing the trajectory information of the $V-I$ trajectory and reducing the discrimination.

- 2) *Feature Extractor Network, Fea(·)*: $V-I$ trajectories usually contain much information in load identification, but valuable features must be extracted for analysis. The task of Fea(·) is to remove high-dimensional feature vectors h from the raw $V-I$ trajectory data x , which may contain information related to different devices or load patterns. This feature extractor network consists of nine 32×32 convolutional layers with batch normalization and LeakyReLU (0.2) between each layer. It maps x to the representation vector $h = \text{Fea}(x) \in R^{D_F}$. x and x' are fed to the feature extractor separately, resulting in a pair of representation vectors. h is normalized to the unit hypersphere in R^{D_F} ($D_F = 2048$ in this article).
- 3) *Projection Network, Proj(·)*: The role of Proj(·) in load identification includes feature dimensionality reduction, feature space normalization, distance metric, and feature representation enhancement, all of which help to improve the performance and robustness of load identification models. We map h to the vector $z = \text{Proj}(h) \in R^{D_P}$. Proj(·) is instantiated as an MLP with a single hidden layer of size 2048 and an output vector of size $D_P = 128$. We again normalize the output of this network to lie on the unit hypersphere, which enables the use of the inner product to measure distances in the projected space. We discard Proj(·) at the end of contrastive training.
- 4) *MLP Classifier, MLP(·)*: In the second phase, we train the MLP(·) for classification using features obtained using Fea(·) that have been trained in the first phase. The vectors obtained in supervised contrastive learning have a higher number of bits than the number of classes N ; therefore, the MLP(·) searches for the best mapping between those vectors and the N -dimensional vectors. The task of MLP(·) is to map the feature vectors obtained from Fea(·) to load types or device classes. This step is critical to load identification as it will match features to known loads to determine which category the observed load belongs to.
- 5) *AutoEncoder, AE(·)*: AE(·) is trained with $V-I$ trajectories of known devices, placed before Fea(·). It can recover the $V-I$ trajectories of the known devices, while the $V-I$ trajectories of the unknown devices will have a larger error from the original $V-I$ trajectories after the AE(·), making the difference between the $V-I$ trajectories of the known and unknown devices even larger to allow the model to better distinguish between known and unknown appliances.

D. Supervised Contrastive Loss

Supervised contrastive loss is a supervised extension of self-supervised loss [42] that is very effective in many downstream tasks. In NILM, $V-I$ trajectory data can be used to learn feature representations by supervised contrastive loss to better capture load behaviors and patterns. We first introduce the self-supervised contrastive loss in self-supervision and then, describe how it is extended to the fully supervised case.

In self-supervised contrastive learning, this is done by randomly sampling N sample pairs, each including an anchor and a positive sample. These sample pairs are used to train the feature extractor network. In NILM, where the anchor and the positive sample correspond to similar electrical devices, these sample pairs can be represented as $V-I$ trajectory data pairs. For N randomly sampled sample pairs, $\{x_k, y_k\}_{k=1, \dots, N}$, the corresponding batch for training consists of $2N$ pairs, $\{x_\ell, y_\ell\}_{\ell=1, \dots, 2N}$, where $\tilde{x}_{2k}$ and $\tilde{x}_{2k-1}$ are the pair of the positive sample corresponding to x_k ($k = 1, \dots, N$), and $\tilde{y}_{2k-1} = \tilde{y}_{2k} = \tilde{y}_k$. The remaining $(2N - 2)$ samples are called negative sample pairs. Then, the self-supervised contrastive learning loss is defined as follows:

$$\mathcal{L}^{\text{self}} = \sum_{i \in I} \mathcal{L}_i^{\text{self}} = - \sum_{i \in I} \log \frac{\exp(z_i \cdot z_{j(i)})/\tau}{\sum_{a \in A(i)} \exp(z_i \cdot z_a)/\tau} \quad (1)$$

where $z_\ell = \text{Proj}(\text{Fea}(\tilde{x}_\ell)) \in R^{D_P}$, the $\cdot$ symbol denotes the inner (dot) product, and $\tau \in R^+$ is the scalar temperature parameter. $i \in I = \{1, \dots, 2N\}$ is the index of the augmented sample and $A(i) \equiv I \setminus i$. The index $j(i)$ denotes a positive sample of the anchor i , and the $(2N - 2)$ indices ($\{k \in A(i) \setminus \{j(i)\}\}$) represent the negative samples.

For the fully-supervised scenario, the loss in (1) cannot make use of the label information. To utilize the label information, the following supervised contrastive loss is proposed:

$$\mathcal{L}^{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p)/\tau}{\sum_{a \in A(i)} \exp(z_i \cdot z_a)/\tau} \quad (2)$$

where $P(i) \equiv \{p \in A(i): \tilde{y}_p = \tilde{y}_i\}$ consists of all the positive indices of the anchor in a base called $|P(i)|$. As shown in (2), for each anchor, all other samples with the same label are considered positive. Using supervised contrastive learning, we can train the Fea(·) to learn compact embedding for each class. We give graphical interpretations of self-supervised contrastive learning and supervised contrastive learning as shown in Fig. 3.

Three types of devices, CFL, WM, and Fridge, are illustrated in Fig. 3. The left half is self-supervised contrastive learning, which is based on the idea of closing the distance between an anchor point and its positive samples and closing the distance between the anchor point and the other negative samples of that anchor point. Positive sample pairs are made up of two subsamples obtained by data augmentation of one sample, one of which serves as the anchor point, the other is his positive sample, and the remaining samples are all considered negative samples. For the sake of esthetics and to be able to see that the positive samples are obtained by data augmentation, the $V-I$ trajectories are not all pictures after data augmentation but are replaced by the original

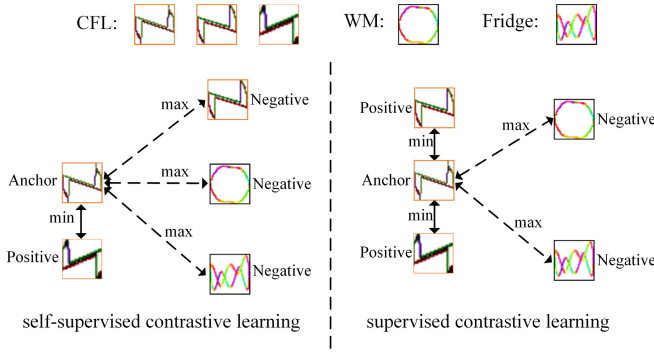


Fig. 3. Illustration of self-supervised contrastive learning and supervised contrastive learning.

$V-I$ trajectories. In the figure, a $V-I$ trajectory of CFL is used as an anchor point, its $V-I$ trajectory after data enhancement is used as its positive sample, and another $V-I$ trajectory of CFL and $V-I$ trajectories of WM and Fridge are used as negative samples. Whereas supervised contrastive learning allows multiple positive samples for an anchor point, the positive and negative correlation between the samples is shown by labeling, i.e., all samples belonging to the same class as the anchor point can be considered as positive samples, and the remaining ones are considered as negative samples. Thus, it can be done to bring the distance between the same class closer and the distance between different classes farther. So all the $V-I$ trajectories belonging to the same class as CFL in the figure can be regarded as positive samples and the remaining $V-I$ trajectories as negative samples.

E. Inference

We use the percentile thresholding (PT) method to compute thresholds to distinguish between known and unknown appliances. PT is intended to separate known appliances from unknown appliances when used in unknown appliance recognition scenarios. The thresholds are set by selecting the appropriate percentile when training the model. We extract the logit before the softmax layer when training the MLP on the training set, ensuring that 99% of the training $V-I$ trajectories are identified as known to obtain the threshold τ . The prediction results are adjusted during testing using the thresholds obtained. Initially, unknown devices were categorized into one of the categories of known devices, and then compared to the thresholds for the corresponding categories based on their maximum logit values, with those above the thresholds being labeled as known and those below the thresholds being labeled as unknown. We refer to this method as threshold-based prediction filtering (TBPF), and this method can be used to increase the confidence of predictions.

III. EXPERIMENTS AND RESULTS

In this section, we first introduce the datasets and evaluation metrics used. Second, we conduct experiments on unknown appliance detection and multiple unknown appliance detection on the PLAID and WHITED datasets and compare them with existing algorithms; the results are discussed and analyzed. Our experiments were implemented using Pytorch 1.8 and

Python 3.6, Intel¹ Xeon¹ W-2133 CPU, 3.60 GHz, 64.00 GB RAM, and NVIDIA Quadro P2000. To exclude appliances from power system fluctuations and noise effects at different times, we set T to 10 to avoid misclassification of the $V-I$ trajectory. The size of the $V-I$ trajectory was set to 32×32 .

A. Data Preparation and Metrics

To assess our proposed algorithm's performance, we utilized two publicly accessible datasets featuring high-frequency submetered current and voltage signals: PLAID, released in 2017 [43], and WHITED [44]. The PLAID dataset comprises current and voltage measurements from various household appliances in Pittsburgh, PA, USA. The 2017 release includes data from 82 appliance instances across 11 appliance types in nine households, totaling 1793 records sampled at 30 kHz. The WHITED dataset records the initial 5 s of voltage and current startup measurements for appliances sampled at 44.1 kHz. WHITED contains 1339 records representing 54 different appliances, offering single-phase measurements for both domestic and industrial use. These records originate from locations in Germany, Austria, Indonesia, and Canada.

Commonly used metrics for evaluating algorithm performance in identifying both unknown and known appliances include the $F1$ -score, F -macro (the average $F1$ -score across all appliances) [14], and area under the ROC curve (AUROC) [45]. The $F1$ -score offers a practical measure of classification accuracy by calculating a weighted average of precision and recall. In open-set scenarios like unknown appliance detection, achieving an $F1$ -score typically involves adding a class to the open set and searching for an appropriate threshold. AUROC, on the other hand, is a metric that does not require a calibration process. It assesses the tradeoff between true positive and false positive rates across various decision thresholds [46].

To validate the proposed algorithm's effectiveness in real-world appliance usage scenarios, we employ a two-step validation process. Initially, we present the results involving the introduction of a single unknown device. Then, in the second step, the results of introducing multiple unknown devices are presented and compared with the literature.

B. Unknown Devices Detection Experiment Results

1) *Experimental Results on PLAID Dataset:* For unknown appliance detection, we train the model using the $V-I$ trajectories of N known appliances and then evaluate its performance on $N + 1$ appliance types. The performance of the method is first evaluated on the PLAID dataset in the presence of only one unknown appliance recognition task, and the performance of the model is reflected based on the F -macro and AUROC scores. The training set comprises $V-I$ trajectories from ten appliances, while the test set includes data from 11 appliances. In our experiments, we adopted cross-validation to designate one appliance as the unknown appliance at a time. This process of training and testing was repeated a total of 11 times. To demonstrate the effectiveness of our proposed algorithm, we compare its experimental results with the latest algorithms.

¹Registered trademark.

TABLE I
F1-SCORE AND AUROC SCORES FOR UNKNOWN DEVICE DETECTION USING DIFFERENT ALGORITHMS ON THE PLAID DATASET

Unknown	Methods	AC	CFL	Fan	Fridge	Hairdryer	Heater	ILB	Laptop	Microwave	Vacuum	WM	F-macro	AUROC
AC	Ours	0.9213	1	0.8571	0.6	0.8696	1	0.8462	0.8333	0.9778	0.9333	0.8571	0.8814	0.9413
	ConOSR	0.8564	1	0.6182	0.5185	0.7925	0.9412	0.8889	0.9231	0.9545	0.8571	0.8571	0.8371	0.8906
	SLCPL	0.4428	0.9778	0.4651	0.1972	0.4948	1	0.8	0.9524	1	0.9333	0.6667	0.7209	0.8929
	ARPL	0.6420	0.9302	0.4167	0.1622	0.84	0.8	0.8235	0.9091	0.9778	0.875	0.5714	0.7225	0.8753
CFL	Ours	0.9500	0.8820	0.8333	0.8	0.8095	1	0.9286	0.4912	1	0.9333	0.8571	0.8623	0.8896
	ConOSR	0.8947	0.5648	0.8947	0.8750	0.9130	0.8	0.9286	0.2236	0.9545	0.7692	0.8571	0.7887	0.6263
	SLCPL	0.7241	0.1162	0.9524	0.5	0.9804	1	0.9655	0.1787	1	0.9333	0.7692	0.7382	0.6971
	ARPL	0.9268	0.3127	0.7568	0.6154	0.9583	0.9412	0.8485	0.1909	0.9778	0.9333	0.75	0.7465	0.6444
Fan	Ours	0.8421	0.9767	0.9351	0.7143	0.8889	1	0.9286	0.8333	0.9778	0.8571	0.8571	0.8919	0.9514
	ConOSR	0.6545	1	0.8926	0.6364	0.9583	0.9412	0.8966	0.8333	0.9545	0.8571	0.8571	0.8620	0.9185
	SLCPL	0.3559	1	0.6358	0.5	0.6024	0.9412	0.9655	0.9	0.9388	0.9333	0.5455	0.7562	0.8518
	ARPL	0.5	0.9091	0.6627	0.6154	0.8148	0.8	0.3256	0.8837	0.9545	0.9333	0.75	0.7408	0.8496
Fridge	Ours	0.9268	0.9767	0.85	0.8358	0.8636	1	0.7692	0.8947	0.9545	0.9333	0.75	0.8868	0.9218
	ConOSR	0.68	0.9767	0.8780	0.7487	0.8889	0.9412	0.8387	0.8333	0.9545	0.8571	0.7778	0.8523	0.9145
	SLCPL	0.5122	0.9778	0.8077	0.4202	0.9434	0.9412	0.8	0.9524	0.8679	0.9333	0.7692	0.8114	0.8635
	ARPL	0.6452	0.9302	0.52	0.2975	0.8980	0.8	0.7179	0.9091	0.9778	0.8571	0.7277	0.7114	0.7784
Hairdryer	Ours	0.8947	0.9524	0.9091	0.7143	0.9363	0.6429	0.9655	0.9231	0.9778	0.8571	0.9333	0.8824	0.9606
	ConOSR	0.95	0.9767	0.7143	0.8750	0.8784	0.4324	0.9286	0.8	0.9545	0.7692	0.7692	0.8226	0.9010
	SLCPL	0.5060	0.9778	0.4419	0.625	0.5876	0.4865	0.7368	0.7692	0.6389	0.9333	0.6667	0.67	0.9199
	ARPL	0.8889	0.9302	0.5417	0.7143	0.8235	0.8	0.4	0.7407	0.9778	0.8571	0.8	0.7704	0.9172
Heater	Ours	0.9756	0.9767	0.8947	0.8	0.9130	0.8673	0.8462	0.8649	1	0.8571	0.8571	0.8957	0.9569
	ConOSR	0.95	1	0.8947	0.8	0.7925	0.8387	0.9286	0.95	0.9545	0.8571	0.8571	0.8930	0.9500
	SLCPL	0.913	0.9778	0.6774	0.7059	0.5682	0.4074	0.9655	0.9302	0.9020	0.9333	0.6667	0.7861	0.8743
	ARPL	0.9268	0.9302	0.5417	0.7143	0.7097	0.7020	0.7368	0.9091	0.9778	0.9333	0.8	0.8074	0.8965
ILB	Ours	0.8333	0.9767	0.7429	0.8	0.9130	1	0.8991	0.8333	0.9545	0.9333	0.9333	0.8927	0.9718
	ConOSR	0.9231	1	0.5217	0.9412	0.8889	0.9412	0.8138	0.8333	0.9778	0.8571	0.8571	0.8687	0.8870
	SLCPL	0.8889	0.9778	0.2745	0.7143	0.9615	0.9412	0.3420	0.9524	1	0.8571	0.2222	0.7393	0.8612
	ARPL	0.9524	0.9302	0.2207	0.6154	0.88	0.9412	0.39	0.9091	0.9778	0.9333	0.75	0.7727	0.7844
Laptop	Ours	0.8947	0.75	0.8649	0.875	0.9583	1	0.9286	0.8949	1	0.8571	0.5385	0.8693	0.9336
	ConOSR	0.8947	0.9362	0.8947	0.8750	0.8636	0.9412	0.9286	0.8683	0.9545	0.8571	0.3	0.8467	0.9432
	SLCPL	0.9767	0.2211	0.9333	0.6154	0.7246	1	0.9655	0.2185	0.8846	0.9333	0.8	0.7521	0.6816
	ARPL	0.9756	0.2973	0.7429	0.8	0.8077	0.9412	0.7368	0.4792	0.9778	0.9333	0.2857	0.7252	0.7919
Microwave	Ours	0.8947	0.9524	0.9231	0.8	0.8889	1	0.9655	0.8947	0.9463	0.8571	0.8571	0.9073	0.9896
	ConOSR	0.8649	0.9268	0.8	0.9412	0.9362	1	0.9286	0.9231	0.9424	0.8571	0.8571	0.9070	0.9782
	SLCPL	0.2979	0.9565	0.6364	0	0.5783	0.8	0.9286	0.9268	0.4893	0.6667	0	0.5709	0.8159
	ARPL	0.8	0.9302	0.7317	0.875	0.6557	0.6667	0.8235	0.7843	0.8632	0.5556	0.8571	0.7766	0.9491
Vacuum	Ours	0.9756	0.9767	0.9231	0.8	1	1	0.9655	0.9500	0.9778	0.9068	0.7692	0.9313	0.9888
	ConOSR	0.75	0.9524	0.8947	0.9412	0.8889	0.9412	0.9286	0.8649	0.9545	0.7758	0.7692	0.8783	0.9489
	SLCPL	0.4421	0.9778	0.9130	0.6667	0.9615	1	0.9655	0.9524	1	0.4043	0.6667	0.8136	0.8752
	ARPL	0.8095	0.9545	0.7778	0.7368	0.8636	0.9412	0.7568	0.9091	0.9778	0.8684	0.8571	0.8593	0.9778
WM	Ours	0.8	0.9524	0.7273	0.5556	0.8636	1	0.8462	0.6667	0.9048	0.9333	0.6889	0.8126	0.8206
	ConOSR	0.85	0.9767	0.9268	0.3810	0.9130	0.9412	0.9655	0.7755	0.9545	0.8571	0.5714	0.8284	0.8589
	SLCPL	0.4516	1	0.7	0.4615	0.9362	0.875	0.8889	0.7234	1	0.5455	0.4912	0.7339	0.7795
	ARPL	0.8261	0.9545	0.6522	0.2745	0.8095	0.9412	0.7568	0.7692	0.9778	0.8571	0.3077	0.7388	0.6663

The experimental results are shown in Table I. The first column signifies appliances treated as unknown appliances. The corresponding rows display the $F1$ -score for known and unknown appliances. The last two columns represent the F -macro and AUROC scores. Cells highlighted in orange signify the detection results for unknown appliances. The F -macro and AUROC scores in Table I demonstrate that our proposed model achieves the best overall performance on the PLAID dataset, surpassing open-set recognition with supervised contrastive learning (ConOSR) [31], adversarial reciprocal points learning for open-set recognition (ARPL) [47], and spatial location constraint prototype loss for open-set recognition (SLCPL) [48]. Only ConOSR has the best performance on the washing machine, on the other devices our model is the highest on both F -macro and AUROC. It can be seen that the proposed algorithm achieves good results for both unknown device detection (orange cells) and known device detection (row indexing). This is because the supervised contrastive learning algorithm closes the distance between appliances of the same kind and expands the distance between appliances of different kinds. We also add an AE to reconstruct the $V-I$ trajectories to further reduce the similarity between known and unknown $V-I$ trajectories.

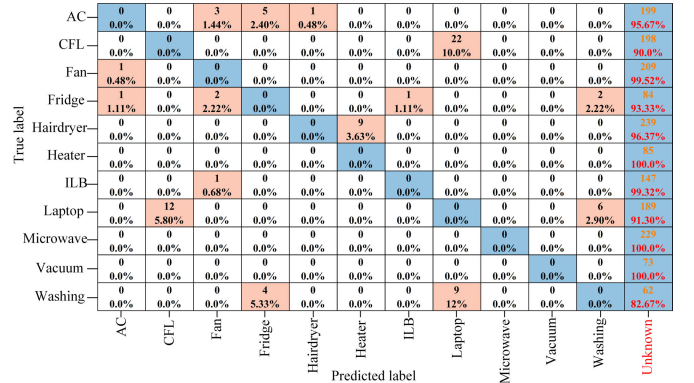


Fig. 4. Confusion matrix when leaving one device out cross-validation is used on the PLAID dataset.

Fig. 4 presents the count of labels per load type, indicating accurate predictions and instances of confusion with other loads. The *true label* on the left signifies individual treatment of these loads as known, while the *predicted label* shows the corresponding predictions, with each row corresponding to one experiment. When considering the CFL as an unknown load, the partial $V-I$ trajectory of the CFL is more similar to the

TABLE II
F1-SCORE AND AUROC SCORES FOR UNKNOWN DEVICE DETECTION USING DIFFERENT ALGORITHMS ON THE WHITED DATASET

Unknown	Methods	Ap1	Ap2	Ap3	Ap4	Ap5	Ap6	Ap7	Ap8	Ap9	Ap10	Ap11	Ap12	F-macro	AUROC
Ap1	Ours	0.9722	0.9565	0.9474	0.9333	1	1	1	0.9714	1	1	1	1	0.9817	0.9995
	ConOSR	0.9091	0.9091	0.6667	0.8571	1	0.9630	0.9714	0.9714	1	1	0.9333	0.9333	0.9262	0.9962
	SLCPL	0.6538	1	1	1	0.3478	1	1	0.8571	1	1	1	1	0.9049	0.9878
	ARPL	0.5234	0.9565	0.9474	0.6667	0.2927	0.7368	0.9730	0.9412	0.6829	0.5	0.6667	0.875	0.7302	0.8703
Ap2	Ours	1	0.96	0.8889	1	1	1	0.9714	0.9714	0.9630	1	1	1	0.9796	0.9959
	ConOSR	0.9231	0.8511	0.8235	0.8571	0.8571	0.88	0.9412	0.9091	0.9231	1	1	0.8571	0.9019	0.9931
	SLCPL	0.9333	0.6517	1	0.9333	1	1	1	1	0.4828	1	1	1	0.9168	0.8829
	ARPL	0.875	0.6	0.8421	0.7692	0.7692	0.9333	0.9091	0.9091	0.4308	0.5	0.6667	0.8571	0.7551	0.6392
Ap3	Ours	1	0.9565	0.9412	0.8889	1	1	1	0.9412	1	1	1	1	0.9773	0.9923
	ConOSR	0.88	0.9091	0.7843	0.6667	0.7692	1	0.9091	0.9714	0.9630	0.8	1	0.8571	0.8758	0.9596
	SLCPL	0.5957	1	0.6885	1	1	1	1	1	1	1	1	1	0.9404	0.8142
	ARPL	0.6829	0.8	0.75	0.9474	0.5455	0.9630	0.9730	0.875	0.6829	0	0.6667	0.8571	0.7286	0.9210
Ap4	Ours	1	1	0.8889	0.9722	1	1	1	0.9714	1	1	0.9333	1	0.9805	0.9975
	ConOSR	0.9231	0.9091	0.3333	0.8642	0.8571	0.9333	0.9714	0.875	1	1	0.9333	0.9333	0.8778	0.9960
	SLCPL	1	1	0.3774	0.6667	1	1	0.9474	1	1	1	1	1	0.9159	0.9880
	ARPL	0.8235	0.9565	0.8421	0.8873	0.5455	0.7692	0.9	0.875	0.7	0.5	0.6667	0.8571	0.7769	0.9624
Ap5	Ours	0.9630	0.9565	0.5714	0.8571	0.8605	1	1	0.9091	0.9231	1	1	1	0.9201	0.8681
	ConOSR	0.9630	0.8571	0.6667	0.6667	0.8945	0.9333	0.9630	0.9714	0.9630	0.9091	0.8571	0.9333	0.8815	0.8494
	SLCPL	1	1	0.2222	1	0.3519	1	1	1	1	1	1	1	0.8812	0.8886
	ARPL	0.9231	0.8	0.5294	0.4	0.8605	0.8571	0.88	0.9444	0.5714	0	0.6667	0.8571	0.6908	0.9486
Ap6	Ours	0.88	1	0.8889	1	1	0.9677	1	1	1	1	1	0.9333	0.9725	0.9969
	ConOSR	0.9231	0.9091	0.5714	0.6667	0.8571	0.8867	0.9630	0.9091	1	1	0.9333	0.8571	0.8731	0.9837
	SLCPL	0.7368	1	1	1	1	0.9024	0.9032	1	0.9333	1	1	0.9412	0.9514	0.9852
	ARPL	0.7179	0.8	0.9474	0.5455	0.5455	0.4286	1	0.9730	0.2947	0.5	0.6667	0.8571	0.6897	0.8383
Ap7	Ours	1	1	0.9474	1	1	1	0.9859	1	0.9714	1	1	1	0.9921	0.9964
	ConOSR	0.9231	0.8	0.1818	0.6667	0.9333	0.88	0.8092	0.9714	0.9091	0.5	0.9333	0.9333	0.7868	0.9944
	SLCPL	1	0.9231	1	0.7692	1	0.6512	0.7521	1	0.8571	1	0.8421	1	0.8996	0.9695
	ARPL	0.8235	0.4444	0.8235	0.6667	0.8571	1	0.5370	0.9231	0.875	0.5	0.5926	0.8	0.7369	0.8917
Ap8	Ours	0.88	1	0.9474	0.9333	1	1	1	0.9091	0.9714	1	1	1	0.9701	1
	ConOSR	0.88	0.9565	0.75	0.6667	0.8571	0.88	0.9412	0.7059	0.875	0.9630	1	0.9333	0.8674	1
	SLCPL	1	1	0.9524	1	1	0.8235	1	0.8462	0.9730	1	1	1	0.9663	1
	ARPL	0.875	0.9565	0.8889	0.6667	0.5455	0.9630	0.9474	0.8571	0.9091	0.9333	1	0.8571	0.8666	0.9942
Ap9	Ours	1	0.9565	0.75	1	1	1	1	0.9714	0.9302	1	1	1	0.9673	1
	ConOSR	0.9231	0.9091	0.3333	0.5455	0.8571	0.9630	0.9091	0.9714	0.7339	0.9630	0.6667	0.9333	0.8090	1
	SLCPL	1	1	1	1	1	1	1	0.7826	0.4615	0.6087	1	1	0.9044	0.9806
	ARPL	0.875	0.8571	1	0.6667	0.6667	1	1	0.9091	0.3607	0.4444	0.5	0.8571	0.7614	0.7133
Ap10	Ours	0.9630	1	0.9474	1	0.9333	1	1	0.9714	1	0.9524	1	1	0.9806	1
	ConOSR	0.9630	0.8	0.1818	0.6667	0.9333	0.9630	0.9714	0.9714	0.9630	0.7619	1	0.8571	0.8361	0.9981
	SLCPL	0.9333	0.7059	1	1	1	0.9655	1	0.9714	0.8	0.6885	1	1	0.9221	0.9983
	ARPL	0.875	0.5806	1	0.6667	0.5455	1	1	0.875	0.5490	0.5075	0.5	0.6667	0.7305	0.8923
Ap11	Ours	0.9231	0.9565	1	1	1	0.7826	0.9714	1	1	1	0.8511	1	0.9571	0.8436
	ConOSR	0.88	0.7368	0.5455	0.8571	0.9630	0.7556	0.9412	0.9630	0.8	1	0.7080	0.9333	0.8403	0.7864
	SLCPL	1	1	1	1	1	0.4186	1	1	1	1	0	1	0.8682	0.7159
	ARPL	0.875	0.8	0.5455	0.6667	0.6667	0.6545	0.9091	0.6512	0	0.6667	0.4810	0.8571	0.6478	0.8628
Ap12	Ours	1	1	0.9474	1	1	1	0.9714	1	1	1	1	0.9756	0.9912	0.9990
	ConOSR	0.9630	1	0.3333	0.7692	0.9630	0.9412	0.8387	0.9630	0.9091	0.9333	0.9333	0.7692	0.8597	0.9964
	SLCPL	0.9655	1	1	1	1	1	1	0.6829	1	1	1	0.7879	0.9530	0.9785
	ARPL	0.9032	0.8	0.9474	0.4	1	0.9730	0.9091	0.4444	0	0.6667	0.8571	0.5075	0.7007	0.6264

partial $V-I$ trajectory of the laptop. It is easy to confuse the two in the unknown identification scenario. When a washing machine is considered as an unknown load, it is easy to be mistaken for a fridge and laptop. Both fridge and washing machine are type II loads with multiple operating states, and the key requirements of the washing machine damper are the same as those of the fridge damper, which both work by motor drive. Consequently, their $V-I$ trajectories can be easily misinterpreted. By analyzing the last column of the confusion matrix, we can determine that the algorithm achieves an average accuracy rate of 98.62% in identifying unknown loads.

2) *Experimental Results on WHITED Dataset:* To demonstrate that our model applies to more than just the PLAID dataset, the proposed model is further tested on the WHITED dataset. The WHITED dataset consists of 54 devices. To show

the recognition performance more clearly, 12 loads were selected for the experiment. Replace the fixed 12 appliances with the following abbreviations: Charger (Ap1), Fan (Ap2), Game Console (Ap3), Hairdryer (Ap4), Kettle (Ap5), LED-light (Ap6), LightBulb (Ap7), Microwave (Ap8), Mixer (Ap9), PowerSupply (Ap10), Toaster (Ap11), and TV (Ap12) [27]. As in the previous experiments, one load was chosen as the unknown load at a time, and the remaining 11 loads were trained as the known loads as the training set; 12 loads were used in the test, and the experiment was repeated 12 times. According to the last two columns of Table II, it can be seen that our proposed algorithm also outperforms the different algorithms, especially in items higher than other algorithms. When comparing the results to those of the PLAID dataset, it's clear that the algorithm's performance improvement is more

TABLE III

F1-SCORE AND AUROC SCORES FOR MULTIPLE UNKNOWN DEVICES DETECTION USING DIFFERENT ALGORITHMS ON THE PLAID DATASET

Unknown	Methods	AC	CFL	Fan	Fridge	Hairdryer	Heater	ILB	Laptop	Microwave	Vacuum	WM	Unknown	F-macro	AUROC
AC Heater Vacuum	Ours	-	0.9767	0.8571	0.9268	0.5517	-	0.8936	0.9655	0.9268	-	0.9778	0.9603	0.8929	0.9539
	ConOSR	-	0.9767	0.7692	0.5574	0.4348	-	0.6765	0.9655	0.8333	-	0.9545	0.8933	0.7846	0.9245
	SLCPL	-	1	0.6154	0.2927	0.3871	-	0.3137	0.4118	0.7778	-	0.5974	0.3807	0.5307	0.8746
	ARPL	-	0.9302	0.3	0.5714	0.0909	-	0.7556	0.6512	0.9091	-	0.9778	0.8182	0.6671	0.906
CFL Fan Vacuum	Ours	0.8710	-	-	0.9231	0.9091	0.7843	1	0.9286	0.6992	-	1	0.9345	0.8944	0.9172
	ConOSR	0.7826	-	-	0.7692	0.8	0.8780	0.9412	0.8889	0.6030	-	0.9778	0.8694	0.8345	0.8439
	SLCPL	0.2485	-	-	0.6667	0.6364	0.4237	0.9474	0.4	0.1634	-	0.7931	0.1473	0.4918	0.7814
	ARPL	0.5538	-	-	0.75	0.5263	0.8571	0.9412	0.3457	0.1743	-	0.8077	0.5907	0.6163	0.7382
Fridge Hairdryer Microwave	Ours	0.9524	1	0.4103	-	-	0.8636	0.72	0.8571	-	0.9268	0.9333	0.9584	0.8469	0.9548
	ConOSR	0.7170	1	0.5833	-	-	0.6207	0.5714	0.8667	-	0.9756	0.8571	0.9396	0.7924	0.9454
	SLCPL	0.1193	0.9778	0.8571	-	-	0.3390	0.8421	0.8571	-	0.6667	0.5455	0.4450	0.6277	0.8103
	ARPL	0.3784	0.9302	0.25	-	-	0.4615	0.5517	0.5	-	0.9091	0.5882	0.7975	0.5963	0.9087
ILB Laptop Vacuum	Ours	0.95	0.9524	0.6667	0.9524	0.8	0.8889	-	-	1	-	0.9778	0.9758	0.9071	0.9705
	ConOSR	0.7391	0.8085	0.2703	0.4828	0.875	0.9362	-	-	0.9412	-	0.9778	0.8962	0.7697	0.9364
	SLCPL	0.6032	0.2189	0.7692	0.2234	0.6667	0.4898	-	-	1	-	0.6765	0.1496	0.5330	0.7264
	ARPL	0.9048	0.4681	0.1308	0.2639	0.6154	0.7667	-	-	0.9412	-	0.9545	0.5630	0.6232	0.7828
Fan Microwave WM	Ours	0.7692	1	-	0.36	0.8889	1	0.9032	0.7234	-	0.8571	-	0.9321	0.8260	0.9238
	ConOSR	0.5769	1	-	0.2667	0.8889	1	0.8966	0.76	-	0.8571	-	0.9098	0.7951	0.9219
	SLCPL	0.1176	0.9778	-	0.625	0.4065	0.9412	0.6222	0.7843	-	0.8571	-	0.3323	0.6293	0.8754
	ARPL	0.3922	0.9302	-	0.2609	0.7755	0.8	0.3182	0.7273	-	0.7692	-	0.8050	0.6421	0.8911

TABLE IV

F1-SCORE AND AUROC SCORES FOR MULTIPLE UNKNOWN DEVICES DETECTION USING DIFFERENT ALGORITHMS ON THE WHITED DATASET

Unknown	Methods	Ap1	Ap2	Ap3	Ap4	Ap5	Ap6	Ap7	Ap8	Ap9	Ap10	Ap11	Ap12	Unknown	F-macro	AUROC
Ap1 Ap2 Ap10	Ours	-	-	0.8889	0.8571	0.5185	1	1	0.9714	0.9655	-	1	1	0.9429	0.9144	0.9820
	ConOSR	-	-	0.1818	0.9333	1	0.88	0.9412	0.9412	0.9630	-	0.8	0.9333	0.9418	0.8516	0.9868
	SLCPL	-	-	1	0.8571	0.3636	0.9032	1	0.5806	0.4667	-	1	1	0.6561	0.7827	0.9304
	ARPL	-	-	1	0.8571	0.3030	1	1	0.8649	0.4667	-	0.5	0.7692	0.8094	0.7570	0.8604
Ap2 Ap4 Ap7	Ours	0.9231	-	0.7619	-	0.9333	0.9333	-	1	0.8293	1	1	0.9333	0.9574	0.9272	0.9785
	ConOSR	0.9630	-	0.5714	-	0.8571	0.8571	-	0.9412	0.9412	0.5	0.875	0.8571	0.9454	0.8309	0.9768
	SLCPL	1	-	0.4878	-	0.6667	1	-	0.7347	0.7660	1	0.9412	0.9412	0.8571	0.8395	0.9571
	ARPL	0.9333	-	0.8235	-	0.1918	0.8571	-	0.9231	0.8421	0.5	0.2133	0.8571	0.5625	0.6704	0.8947
Ap5 Ap6 Ap9	Ours	1	0.9565	0.3462	0.9333	-	-	1	1	-	0.9630	1	1	0.9095	0.9109	0.9466
	ConOSR	0.7586	0.9565	0.5333	0.6667	-	-	0.6667	1	-	1	1	0.9333	0.9333	0.8639	0.9349
	SLCPL	0.7179	0.8571	0.2198	1	-	-	1	0.6364	-	0.5957	1	0.6667	0.5825	0.7276	0.9356
	ARPL	0.7778	0.8	0.6667	0.6	-	-	0.4	0.2917	-	0.2435	0.8	0.8571	0.3564	0.5793	0.7922
Ap3 Ap8 Ap12	Ours	1	0.9565	-	0.8235	0.9630	1	0.9714	-	1	1	1	-	0.9735	0.9688	0.9803
	ConOSR	0.7059	0.9091	-	0.1818	1	0.9412	0.9412	-	0.9231	0.9333	0.9333	-	0.8755	0.8344	0.9744
	SLCPL	0.4179	1	-	1	1	1	0.9444	-	0.7568	0.6667	1	-	0.7045	0.8490	0.8878
	ARPL	0.5833	0.5882	-	0.9474	0.9630	0.9730	0.9091	-	0.5385	0.7692	0.8571	-	0.7604	0.7889	0.8499
Ap5 Ap7 Ap11	Ours	0.88	1	0.9333	0.9333	-	0.3889	-	0.9714	1	1	-	0.9333	0.8661	0.8906	0.8666
	ConOSR	0.9231	0.6429	0.6667	1	-	0.2281	-	0.9714	1	1	-	0.9333	0.6845	0.8050	0.7664
	SLCPL	0.9032	0.7059	0.9333	1	-	0.1867	-	0.9189	1	1	-	1	0.5159	0.8164	0.7958
	ARPL	0.8485	0.5714	0.5556	0.8571	-	0.1783	-	0.8485	0.5	0.8	-	0.8571	0.4059	0.6422	0.5595

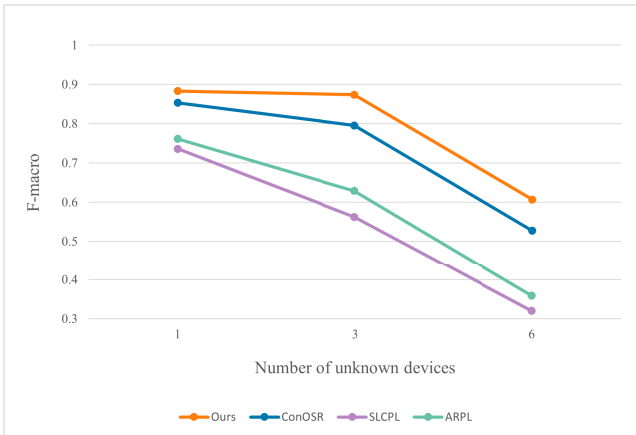


Fig. 6. Changes in algorithm performance as the number of unknown devices on the PLAID dataset increases.

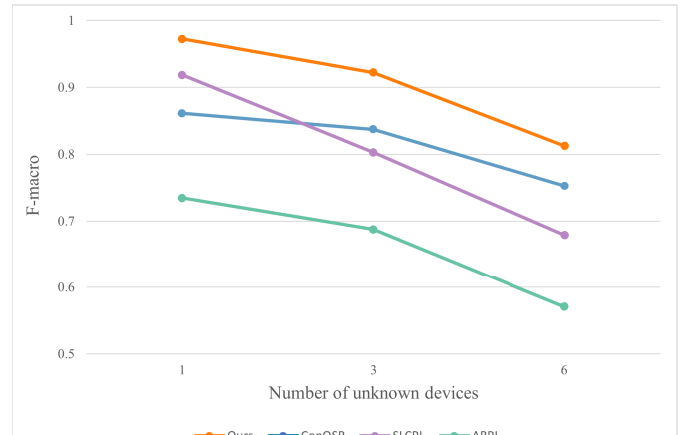


Fig. 7. Changes in algorithm performance as the number of unknown devices on the WHITED dataset increases.

IV. ABLATION STUDY

We conducted ablation experiments on the PLAID and WHITED datasets and tested them on four algorithms:

traditional open-set supervised contrastive learning (ConOSR), open-set supervised contrastive learning with our improved

TABLE V
F-MACRO SCORES FOR ABLATION EXPERIMENTS ON THE PLAID DATASET

Method	AC	CFL	Fan	Fridge	Hairdryer	Heater	ILB	Laptop	Microwave	Vacuum	WM	Average
ConOSR	0.8371	0.7887	0.8620	0.8523	0.8226	0.8930	0.8687	0.8467	0.9070	0.8783	0.8284	0.8532
Ours/AE	0.8709	0.8432	0.8736	0.8586	0.8306	0.8930	0.8796	0.8236	0.8973	0.9147	0.8305	0.8650
Ours/Aug	0.8581	0.8423	0.8840	0.8663	0.8720	0.8945	0.8713	0.8544	0.9038	0.9273	0.8096	0.8712
Ours	0.8814	0.8623	0.8919	0.8868	0.8824	0.8957	0.8927	0.8693	0.9073	0.9313	0.8126	0.8831

TABLE VI
AUROC SCORES FOR ABLATION EXPERIMENTS ON THE PLAID DATASET

Method	AC	CFL	Fan	Fridge	Hairdryer	Heater	ILB	Laptop	Microwave	Vacuum	WM	Average
ConOSR	0.8906	0.6263	0.9185	0.9145	0.9010	0.9500	0.8870	0.9432	0.9782	0.9489	0.8589	0.8925
Ours/AE	0.9555	0.8819	0.9454	0.9611	0.9569	0.9760	0.9680	0.8712	0.9764	0.9804	0.8405	0.9376
Ours/Aug	0.9128	0.7774	0.9326	0.8733	0.9158	0.9151	0.9087	0.9067	0.9795	0.9887	0.8311	0.9038
Ours	0.9413	0.8896	0.9514	0.9218	0.9606	0.9569	0.9718	0.9336	0.9896	0.9888	0.8206	0.9387

TABLE VII
F-MACRO SCORES FOR ABLATION EXPERIMENTS ON THE WHITED DATASET

Method	Ap1	Ap2	Ap3	Ap4	Ap5	Ap6	Ap7	Ap8	Ap9	Ap10	Ap11	Ap12	Average
ConOSR	0.9262	0.9019	0.8758	0.8778	0.8815	0.8731	0.7868	0.8674	0.8090	0.8361	0.8403	0.8597	0.8613
Ours/AE	0.9347	0.9038	0.8821	0.9269	0.8939	0.8967	0.8886	0.8827	0.8672	0.8897	0.8881	0.9285	0.8986
Ours/Aug	0.9593	0.9604	0.9443	0.9457	0.9006	0.9685	0.9552	0.9514	0.9595	0.9660	0.9393	0.9685	0.9515
Ours	0.9817	0.9796	0.9773	0.9805	0.9201	0.9725	0.9921	0.9701	0.9673	0.9806	0.9571	0.9912	0.9725

TABLE VIII
AUROC SCORES FOR ABLATION EXPERIMENTS ON THE WHITED DATASET

Method	Ap1	Ap2	Ap3	Ap4	Ap5	Ap6	Ap7	Ap8	Ap9	Ap10	Ap11	Ap12	Average
ConOSR	0.9962	0.9931	0.9596	0.9960	0.8494	0.9837	0.9944	1	1	0.9981	0.7864	0.9964	0.9628
Ours/AE	0.9997	1	0.9350	1	0.8670	0.9988	0.9978	1	1	0.9990	0.7322	1	0.9608
Ours/Aug	0.9847	0.9743	0.9775	0.9562	0.8912	0.9869	0.9733	0.9965	0.9990	0.9956	0.8619	0.9994	0.9664
Ours	0.9995	0.9959	0.9923	0.9975	0.8681	0.9969	0.9964	1	1	1.0000	0.8436	0.9990	0.9741

data augmentation method added (Ours/AE), open-set supervised contrastive learning with AE added (Ours/Aug) and our proposed method using ConOSR as a baseline.

A. Ablation Experiments on PLAID Dataset

To reduce the contingency, we conducted 11 experiments for each algorithm, each of which contains only one unknown appliance and will not be repeated, finally, the results of the 11 experiments are averaged. Tables V and VI show the F-macro and AUROC scores of the four algorithms. From the two tables, it can be seen that our proposed algorithm obtains the highest scores in both evaluation metrics, and the results of the remaining two algorithms are also stronger than the baseline. It proves that the improvements we made are effective.

B. Ablation Experiments on WHITED Dataset

To further demonstrate our proposed algorithm, ablation experiments were again done on the WHITED dataset. The experimental setup is basically the same as the PLAID dataset. From Tables VII and VIII, it can be seen that our proposed algorithm and the improvements made are basically better than the baseline ones. Once again, the performance of our method is proven.

V. CONCLUSION AND FUTURE WORK

In this article, a reconfiguration-based supervised contrastive learning is proposed for unknown device identification in

NILM. We integrate supervised contrastive learning and AE and improve the data augmentation approach. Using $V-I$ trajectory-signed appliances as input, supervised contrastive learning clusters together cluster points belonging to the same class of appliances in the embedding space while separating appliances of different classes. We change the data enhancement approach so that the $V-I$ trajectory maximizes the preservation of appliance information but is different from the original $V-I$ trajectory. We apply AE to the downstream classification task to reconstruct the semantic representation of the image so that the boundary between known and unknown appliances is more obvious. This further improves the performance of the model. We distinguish known from unknown appliances by assessing the distance between the feature distributions of the test sample and the known appliance sample. The algorithm shows excellent performance in unknown appliance detection. In future work, we aim to enhance our algorithm by incorporating detected unknown appliances into the model. This approach will enable more efficient detection of unknown appliances and expand the practical applications of NILM.

REFERENCES

- [1] D. Alahakoon and X. Yu, "Smart electricity meter data intelligence for future energy systems: A survey," *IEEE Trans. Ind. Informat.*, vol. 12, no. 1, pp. 425–436, Feb. 2016.

- [2] S. S. Hosseini, A. Badri, and M. Parvania, "A survey on mobile energy storage systems (MESS): Applications, challenges and solutions," *Renew. Sustain. Energy Rev.*, vol. 40, pp. 161–170, Dec. 2014.
- [3] U. Doe, "An assessment of energy technologies and research opportunities," in *Quadrennial Technology Review*. Washington, DC, USA: United States Department of Energy, 2015, pp. 12–19.
- [4] A. Ridi, C. Gisler, and J. Hennebert, "A survey on intrusive load monitoring for appliance recognition," in *Proc. 22nd Int. Conf. Pattern Recognit.*, Aug. 2014, pp. 3702–3707.
- [5] G. W. Hart, "Nonintrusive appliance load monitoring," *Proc. IEEE*, vol. 80, no. 12, pp. 1870–1891, Jun. 1992.
- [6] O. Parson, S. Ghosh, M. Weal, and A. Rogers, "Non-intrusive load monitoring using prior models of general appliance types," in *Proc. 26th Conf. Artif. Intell. (AAAI)*, Jul. 2012, pp. 356–362.
- [7] W. Kong, Z. Y. Dong, D. J. Hill, J. Ma, J. H. Zhao, and F. J. Luo, "A hierarchical hidden Markov model framework for home appliance modeling," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 3079–3090, Jul. 2018.
- [8] D. Egarter, V. P. Bhuvana, and W. Elmenreich, "PALDi: Online load disaggregation via particle filtering," *IEEE Trans. Instrum. Meas.*, vol. 64, no. 2, pp. 467–477, Feb. 2015.
- [9] S. Chen, B. Zhao, M. Zhong, W. Luan, and Y. Yu, "Nonintrusive load monitoring based on self-supervised learning," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.
- [10] C. Zhang, M. Zhong, Z. Wang, N. Goddard, and C. Sutton, "Sequence-to-point learning with neural networks for non-intrusive load monitoring," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, 2018, pp. 2604–2611.
- [11] C. Xiong, Z. Cai, S. Liu, J. Luo, and G. Tu, "An improved sequence-to-point learning for non-intrusive load monitoring based on discrete wavelet transform," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–16, 2023.
- [12] J. Lu, R. Zhao, B. Liu, Z. Yu, J. Zhang, and Z. Xu, "An overview of non-intrusive load monitoring based on V-I trajectory signature," *Energies*, vol. 16, no. 2, p. 939, Jan. 2023.
- [13] H. Lam, K. Ting, W. Lee, and G. Fung, "An analytical understanding on voltage-current curve of electrical load," in *Proc. Int. Conf. Electr. Eng. (ICEE)*, 2006, pp. 1–6.
- [14] Y. Liu, X. Wang, and W. You, "Non-intrusive load monitoring by voltage-current trajectory enabled transfer learning," *IEEE Trans. Smart Grid*, vol. 10, no. 5, pp. 5609–5619, Sep. 2019.
- [15] R. de Paula Rodrigues and P. M. da Silva, "Curvature scale space-based signatures for electrical load classification in NILM," *Electr. Eng.*, vol. 103, no. 2, pp. 1239–1252, Apr. 2021.
- [16] Y. Han, K. Li, H. Feng, and Q. Zhao, "Non-intrusive load monitoring based on semi-supervised smooth teacher graph learning with voltage-current trajectory," *Neural Comput. Appl.*, vol. 34, no. 21, pp. 19147–19160, 2022.
- [17] P. Zhang, B. Gao, H. Chen, and Z. Yu, "Appliance recognition using V-I trajectories based on deep learning," in *Proc. IEEE 4th Int. Conf. Power, Intell. Comput. Syst. (ICPICS)*, Jul. 2022, pp. 239–243.
- [18] D. H. Green, A. W. Langham, R. A. Agustin, D. W. Quinn, and S. B. Leeb, "Adaptation for automated drift detection in electromechanical machine monitoring," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 6768–6782, 2023.
- [19] C. Dinesh, B. W. Nettasinghe, R. I. Godaliyadda, M. P. B. Ekanayake, J. Ekanayake, and J. V. Wijayakulasooriya, "Residential appliance identification based on spectral information of low frequency smart meter measurements," *IEEE Trans. Smart Grid*, vol. 7, no. 6, pp. 2781–2792, Nov. 2016.
- [20] H. G. C. P. Dinesh, P. H. Perera, G. M. R. I. Godaliyadda, M. P. B. Ekanayake, and J. B. Ekanayake, "Residential appliance monitoring based on low frequency smart meter measurements," in *Proc. IEEE Int. Conf. Smart Grid Commun. (SmartGridComm)*, Nov. 2015, pp. 878–884.
- [21] H. Rashid, P. Singh, V. Stankovic, and L. Stankovic, "Can non-intrusive load monitoring be used for identifying an appliance's anomalous behaviour?" *Appl. Energy*, vol. 238, pp. 796–805, Mar. 2019.
- [22] S. Welikala, C. Dinesh, R. I. Godaliyadda, M. P. B. Ekanayake, and J. Ekanayake, "Robust non-intrusive load monitoring (NILM) with unknown loads," in *Proc. IEEE Int. Conf. Inf. Autom. Sustainability (ICIAS)*, Dec. 2016, pp. 1–6.
- [23] Q. Zhao, Y. Xu, Z. Wei, and Y. Han, "Non-intrusive load monitoring based on deep pairwise-supervised hashing to detect unidentified appliances," *Processes*, vol. 9, no. 3, p. 505, 2021.
- [24] M. Yu, B. Wang, L. Lu, Z. Bao, and D. Qi, "Non-intrusive adaptive load identification based on Siamese network," *IEEE Access*, vol. 10, pp. 11564–11573, 2022.
- [25] J. Zhang, X. Chen, W. W. Ng, C. S. Lai, and L. L. Lai, "New appliance detection for nonintrusive load monitoring," *IEEE Trans. Ind. Informat.*, vol. 15, no. 8, pp. 4819–4829, Aug. 2019.
- [26] L. De Baets, C. Develder, T. Dhaene, and D. Deschrijver, "Detection of unidentified appliances in non-intrusive load monitoring using Siamese neural networks," *Int. J. Elect. Power Energy Syst.*, vol. 104, pp. 645–653, Jan. 2019.
- [27] Y. Han, K. Li, C. Wang, F. Si, and Q. Zhao, "Unknown appliances detection for non-intrusive load monitoring based on conditional generative adversarial networks," *IEEE Trans. Smart Grid*, vol. 14, no. 6, pp. 4553–4564, 2023.
- [28] A. Gao, F. Mei, J. Zheng, H. Sha, M. Guo, and Y. Xie, "Electricity theft detection based on contrastive learning and non-intrusive load monitoring," *IEEE Trans. Smart Grid*, vol. 14, no. 6, pp. 4565–4580, 2023.
- [29] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult, "Toward open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 7, pp. 1757–1772, Jul. 2013.
- [30] Y. Yang, Y. Ge, B. Li, Q. Wang, Y. Lang, and K. Li, "Multiscenario open-set gait recognition based on radar micro-Doppler signatures," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.
- [31] Y. Kodama, Y. Wang, R. Kawakami, and T. Naemura, "Open-set recognition with supervised contrastive learning," in *Proc. 17th Int. Conf. Mach. Vis. Appl. (MVA)*, Jul. 2021, pp. 1–5.
- [32] S. Vaze, K. Hant, A. Vedaldi, and A. Zisserman, "Generalized category discovery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7482–7491.
- [33] H. Huang, Y. Wang, Q. Hu, and M.-M. Cheng, "Class-specific semantic reconstruction for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4214–4228, Apr. 2023.
- [34] P. Oza and V. M. Patel, "C2AE: Class conditioned auto-encoder for open-set recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2307–2316.
- [35] Y. Kawachi, Y. Koizumi, and N. Harada, "Complementary set variational autoencoder for supervised anomaly detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2018, pp. 2366–2370.
- [36] P. Peng et al., "Imbalanced fault diagnosis by supervised contrastive learning," in *Proc. IEEE 25th Int. Conf. Comput. Supported Cooperat. Work Design (CSCWD)*, May 2022, pp. 689–692.
- [37] Y. Zhang, L. Gao, X. Wen, and H. Wang, "Intelligent fault diagnosis of machine under noisy environment using ensemble orthogonal contractive auto-encoder," *Expert Syst. Appl.*, vol. 203, Oct. 2022, Art. no. 117408.
- [38] R. Wang et al., "Class-aware variational auto-encoder for open set recognition," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 264–269.
- [39] K. He, L. Stankovic, J. Liao, and V. Stankovic, "Non-intrusive load disaggregation using graph signal processing," *IEEE Trans. Smart Grid*, vol. 9, no. 3, pp. 1739–1747, May 2018.
- [40] B. Zhao, K. He, L. Stankovic, and V. Stankovic, "Improving event-based non-intrusive load monitoring using graph signal processing," *IEEE Access*, vol. 6, pp. 53944–53959, 2018.
- [41] J. Luo, Y. Zhang, Y. Wu, Y. Xu, X. Guo, and B. Shang, "A multi-channel contrastive learning network based intrusion detection method," *Electronics*, vol. 12, no. 4, p. 949, Feb. 2023.
- [42] P. Khosla et al., "Supervised contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 18661–18673.
- [43] L. De Baets, C. Develder, T. Dhaene, D. Deschrijver, J. Gao, and M. Berges, "Handling imbalance in an extended plaid," in *Proc. Sustain. Internet ICT Sustainability (SustainIT)*, Dec. 2017, pp. 1–5.
- [44] M. Kahl, A. U. Haq, T. Kriebchaumer, and H.-A. Jacobsen, "WHITED—A worldwide household and industry transient energy data set," in *Proc. 3rd Int. Workshop Non-Intrusive Load Monitor.*, 2016, pp. 1–4.
- [45] C. Geng, S.-J. Huang, and S. Chen, "Recent advances in open set recognition: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3614–3631, Oct. 2021.
- [46] W. Moon, J. Park, H. S. Seong, C.-H. Cho, and J.-P. Heo, "Difficulty-aware simulator for open set recognition," in *Proc. Eur. Conf. Comput. Vis. Berlin, Germany: Springer-Verlag*, 2022, pp. 365–381.

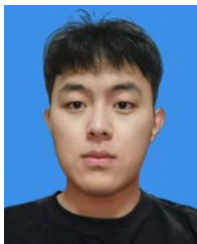
- [47] G. Chen, P. Peng, X. Wang, and Y. Tian, "Adversarial reciprocal points learning for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8065–8081, Nov. 2022.
- [48] Z. Xia, P. Wang, G. Dong, and H. Liu, "Spatial location constraint prototype loss for open set recognition," *Comput. Vis. Image Understand.*, vol. 229, Mar. 2023, Art. no. 103651.



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